Shruthi Mohan

CONTACTS

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SKILLS

Professional Electives: ML, AI algorithms, Linear Algebra, Statistical Probability, DSA, Deep Learning, Image Processing, Speech Processing, NLP, OCR
Programming Languages: C, C++, Java, R, & Python
Big Data Analytics: Hadoop, PySpark
Data Visualization Platforms: Tableau, Plotly Dash, Microsoft Power BI
Database: SQL, PL/SQL, Firebase
Front-End Technologies: HTML, CSS, Javascript & React.js
Python Libraries: BeautifulSoup, Pandas, Numpy, Sklearn, Matplotlib, Flask, Tensorflow
Hackerrank: Python - 4 stars | Java - 2 stars | C - 4 stars
Leetcode: Solved 275+ python problems | Leetcode 75 badge

LEADERSHIP SKILLS

- **Scrum Master** of the project during Infosys Springboard Internship: (Jan'23–Apr'23)
- **Club Head of the Chess Club** at Shiv Nadar University Chennai [founded the club]:(Dec'21-)
- **Head of the Graphics & Design team** at Cosmos - IIT Madras BS Tech Society:(Mar'23-Jun'23)
- **Coordinator of PixelMind AI** event of IIT Madras BS Paradox'23:(May'23)
- Joint Secretary of Cosmos - IIT Madras BS Tech Society: (Jun'23-Sep'23)
- **Team Lead** of Kavach Project- Advanced Fake News Detection (Apr'23)

LANGUAGE

English: Expert | **Hindi**: Intermediate | **Tamil**: Intermediate | **French**: Novice(A1Delf)

PROFILE

Problem-solving enthusiast tackling challenges with AI & Data Science. Eager to learn and contribute to real-world solutions.

EDUCATION

Shiv Nadar University Chennai
B.Tech Artificial Intelligence and Data Science
4th Year (2021-25) | 8.5 CGPA
Indian Institute of Technology Madras (IITM)
B.Sc in Programming and Data Science
Diploma level (2022-25) | 7.2 CGPA
Sri Chaitanya Techno School, Perumbakkam, Chennai
12th grade (2021) | 94%
Padma Seshadri Bala Bhavan Millennium OMR, Chennai
10th grade (2019) | 91%

WORK EXPERIENCE

Part-Time Trainee | Video Yantra Solutions (Apr'23 - Apr'24)
- Developed and deployed ML models to streamline and automate client project workflows, improving efficiency and accuracy.
- Rigorously tested Android applications built with React Native to ensure optimal performance and user experience.
- Gained experience in front-end development using React.js.

INTERNSHIP EXPERIENCE

Software Engineering Intern | Google (May'24 - Aug'24)
- **NLP:** Expanded cache parser coverage for Assistant NLP i18n team through similarity-based matching and encoder retraining.
- **System Optimization & Code Comprehension:** Deepened the team's understanding of the codebase to optimize system utilization.
- **Additional project:** Developed configuration file to streamline i18n LE Analysis.

Student Trainee | Tech Mahindra (May'23 - June'23)
- Developed real-time stock prediction dashboard using LSTM, Plotly Dash.
- Employed deep learning for time series analysis and prediction.
- Built interactive web application with automated data visualization.
- Enhanced skills in machine learning, time series analysis, and web development

Project Intern & Scrum Master | Infosys Springboard (Jan'23 – Apr'23)
- Led Scrum team as Scrum Master to develop an Automatic Dataset Cleaner.
- Used Django backend for data pre-processing & manipulation skills for efficient cleaning using Pandas.
- Created HTML/CSS frontend for user interaction.
- Demonstrated full-stack development and Agile leadership.

Data Science and Business Analytics Intern | The Spark Foundation (Dec'22)
- Honed data cleaning and visualization skills during internship.
- Cleaned the Global Terrorism Dataset using data manipulation techniques.
- Developed an interactive dashboard for data exploration.

PAPER PRESENTATION

Multi-Agent Search using RL:
a) Chennai Institute of Technology (Mar'23)
b) SRM Valliammai College(Sep'23)
Presented a paper that used multi-modal maps and a decentralized search process to address the problem of aerial swarms search with a group of agents using probability and uncertainty. LINK: http://surl.li/ujpkny

CERTIFICATIONS

Forage: Goldman Sachs: Virtual Experience Program Participant Cryptography (Oct'22)
Infosys Springboard:
- Python Programmer (Jan'22)
- Power BI for Business Professionals (Nov'22)
- Multidimensional Data Modeling (Nov'22)
- Regression Analysis (Dec'22)
- Time Series using Python (Dec'22)

IBM-Coursera:
- Data Analysis with Python (Nov'22)
- Excel Basics for Data Analysis (Dec'22)
- Clean Data in SQL using MySQL Workbench (Dec'22)

NPTEL:
- Advanced R Programming for Data Analytics in Business (Oct'24)
- Design and Implementation of Human and Computer Interface (Oct'24)
- Blockchain and its Applications (Apr'24)
- Business Intelligence and Analytics (Apr'24)
- Cloud Computing (Oct'23)
- Python for Data Science (Oct'21)
- Introduction to Machine Learning (Apr'22)

ACHIEVEMENTS

- 2nd place in VIT Chennai's Data Spark ML Hackathon
- 1st place in SRM Valliammai Paper Presentation
- SSN Trophy Women's Chess (2nd place)
- SNU Inter-college Women's chess (1st place)

EXTRACURRICULAR

- **Google Intern Connect:** Volunteered and connected with interns from other product-based companies such as Microsoft.
- **Trinity College of London**: Completed keyboard (2nd level) and Theory of Music (3rd level)

PROJECTS

1) Intent Detection (Dec'24): Leveraged deep learning for intent detection, achieving 70% accuracy. Employed BERT embeddings for rich contextual understanding and fine-tuned neural network with class weights and hyperparameter optimization. Utilized TensorFlow/Keras, scikit-learn, and Transformers for model building and feature engineering.
LINK: https://github.com/shruthimohan03/Intent-Detection

2) Content Engine - Query & Compare PDFs (Nov'24): Developed a PDF-based query engine using LangChain and FAISS for semantic search, enabling efficient retrieval of relevant content from document collections. Retrieval-augmented generation (RAG) principles were applied to enhance user query responses.
LINK: https://github.com/shruthimohan03/content-engine

3) Predict attributes from product image (Oct'24): Leveraged image processing techniques, deep learning and computer vision to predict product attributes from supplier-uploaded images. Combined image features (ViT) and tabular data for robust neural network (Tensorflow) predictions. Addressed missing attribute values and category-specific attributes.
LINK:https://github.com/shruthimohan03/Predict-attributes-from-product-image

4) Text-to-Video (Aug'24-Nov'24): Built a Text-to-Video model using CLIP and Stable Diffusion. Implemented frame interpolation, and transition techniques for image to video. Explored feature extraction models for image analysis.

5) Emotion Recognition for Conversational Data (Dec'23-Mar'24): Successfully trained and evaluated multiple classification models for emotion recognition. Explored various text embedding and classification techniques. Demonstrated proficiency in natural language processing (NLP) and sentiment analysis.
LINK: https://github.com/shruthimohan03/Emotion_Recognition

6) Parallel Database Query System (Dec'23-Mar'24): Designed a high-performance database architecture. Implemented parallel processing for data, algorithms, queries, and inputs using cutting-edge techniques.
LINK: https://github.com/shruthimohan03/Parallel-Database-Architecture

7) Smart India Hackathon: Sentiment Analysis on incoming calls in call centers (Sep'23-Oct'23): In a 6-member team to develop a sentiment analysis-based customer service application—leveraged WSA-BERT to enhance customer interactions and satisfaction. The project was recognized at Shiv Nadar University's internal hackathon.

8) Kavach: Advanced Fake News Detection (Apr'23): Team Lead of a 6-member team to develop a multi-faceted fake news detection system. Proposed browser plugin and chatbot solutions to combat misinformation on web articles and social media. The project was selected in Shiv Nadar University's internal hackathon.

9) AI-based Reinforcement Learning (Sep'22-Oct'22): Based on RL, environments and agents were built based on agent response and behaviors using Python, Java 17, and Gradle. We used simulations to train different rescue agents(ambulance, fire station) to undertake operations effectively.

10)MLOPs-based recommendation system (Jan'23): Created a model that recommends the top ten closely related series based on the user's input of an available series. Using Flask framework for back and front-end development, the Machine Learning model was connected to a dynamic HTML page. The ML model was abstracted using pickle.
LINK: https://github.com/shruthimohan03/Recommendation-Model-using-Flask